Notes: Bernard, Redding, and Schott (AER, 2010)

Multiple-Product Firms and Product Switching

Contents

1 Big picture

- **Question.** How important is product switching inside continuing firms, and what does it reveal about resource reallocation in manufacturing?
- **Core idea.** Reallocation is not only about firm entry and exit. Surviving firms also reallocate resources by adding and dropping products. This within-firm extensive margin is frequent, economically large, and systematically related to firm and firm-product characteristics.
- **Setting.** The paper uses U.S. Census of Manufactures data from 1987, 1992, and 1997. Products are approximately 1,500 five-digit SIC manufacturing categories. Firms can produce one product, multiple products, products in multiple four-digit industries, or products in multiple two-digit sectors.
- **Main facts.** More than one-half of surviving U.S. manufacturing firms change their product mix over a five-year census interval. Firms that switch products account for the overwhelming majority of output. Product adding and dropping together explain a large share of changes in both firm scope and product-level output.
- **Main interpretation.** The facts are consistent with a model in which firms have firm-level productivity and firm-product-specific demand or profitability shocks. More productive firms can profitably carry a wider product range; within a firm, weaker or younger products are more likely to be dropped.
- **Why it matters.** Studies of productivity and output growth that focus only on entry, exit, or changes across firms miss an important reallocation margin: selection of products within firms.

2 Incremental contribution of the paper

- **New unit of analysis.** The paper moves beyond the usual firm-entry and firm-exit margin and studies product entry and exit *within* continuing firms.
- **Comprehensive data.** It uses a longitudinal census dataset that tracks product-level manufacturing output within firms across U.S. Manufacturing Censuses. This is much broader than datasets that observe only a firm's primary industry.
- **Endogenous product selection.** The paper develops a model in which firms choose an endogenous product range. The model extends standard industry dynamics and heterogeneous-firm models by separating firm-level productivity from firm-product demand/profitability.
- **Selection within firms.** The empirical results show that firms drop products in a systematic way: products with lower relative shipments and shorter firm-product tenure are more likely to disappear. This is direct evidence of reallocation inside the firm.
- **Gross reallocation.** The paper documents that gross adding and dropping are large relative to net changes. This mirrors the job-creation/job-destruction literature: a stable aggregate can hide large underlying churn.
- **Limits of the baseline model.** The paper also documents coproduction patterns and M&A modes of product switching. These facts point to richer theories with product complementarities, shared capabilities, and plant transactions.

3 Model structure

3.1 Demand and products

Consumers have CES preferences across products, and each product is itself a CES aggregate of firm varieties. A product is indexed by i, a firm variety by ω, and product-specific demand is affected by a firm-product demand parameter $\lambda_i(\omega)$.

Interpretation: The demand side gives firms market power within products. Product-specific demand or quality differences mean the same firm can have strong demand in one product and weak demand in another.

3.2 Firms, product scope, and productivity

A potential firm pays a sunk entry cost and then observes:

- a firm-level productivity draw φ that applies to all products;
- a vector of firm-product demand draws λ_i that differ across products.

Each product requires a fixed production cost. A firm produces a product only if the product-specific profit covers that fixed cost. Therefore there is a zero-profit cutoff for each product, $\lambda_i^*(\varphi)$. More productive firms have lower cutoffs and can profitably produce more products.

Economic message: Firm scope is endogenous. High-productivity firms produce more products because productivity raises expected revenue in all products, making it easier to cover fixed product costs.

3.3 Dynamics and shocks

The model includes stochastic shocks to firm productivity and firm-product demand. Firm-level productivity shocks shift the profitability of all products at a firm. Product-specific shocks affect one firm-product pair.

- A positive firm-level shock can lead the firm to add products and expand output.
- A negative firm-level shock can lead the firm to drop products and contract.
- A negative firm-product shock makes a particular product more likely to be dropped, even if the firm survives.

Takeaway: The model generates both firm entry/exit and product adding/dropping by continuing firms. It predicts that product switching should be related to both firm characteristics and firm-product characteristics.

4 Core theoretical predictions

1. **Multiple-product firms should be larger and more productive.** Higher productivity lowers the profitability cutoff for producing products and raises firm scope.
2. **Product switching should be pervasive.** Shocks to firm-product demand generate continuous product adding and dropping even among surviving firms.

3. **Product add and drop rates should be positively correlated across products.** Turbulent product markets generate both frequent adds and frequent drops.
4. **Product drops should be scale dependent.** A firm is less likely to drop a product if its shipments of that product are large relative to other firms producing the same product.
5. **Product drops should be age dependent.** A product that a firm has produced for longer is less likely to be dropped.
6. **Net adding should be associated with growth and productivity gains.** Firms that net add products should experience increases in output, employment, and revenue-based productivity; firms that net drop should shrink.

5 Data and measurement

5.1 U.S. Census of Manufactures

The empirical work uses the quinquennial U.S. Census of Manufactures for 1987, 1992, and 1997. Establishments report shipments by detailed SIC categories, and the paper aggregates plant-level product reports to the firm-product-year level.

- **Product:** five-digit SIC category.
- **Industry:** four-digit SIC category.
- **Sector:** two-digit SIC category.
- **Single-product firm:** produces one five-digit product.
- **Multiple-product firm:** produces more than one five-digit product.

The paper focuses on surviving firms when measuring product switching. Entering firms are not counted as adding all their products, and exiting firms are not counted as dropping all products. This keeps the focus on the within-firm margin.

5.2 Product switching definitions

For each surviving firm between census years, the product mix can be unchanged or can involve:

- **Drop only:** the firm drops at least one product and adds none;
- **Add only:** the firm adds at least one product and drops none;
- **Both add and drop:** the firm churns products;
- **None:** the product mix is unchanged.

Interpretation: Because the data are quinquennial, the estimates likely understate true product switching. Product changes within a five-year period that are reversed before the next census are not observed.

6 Empirical strategy

The paper is primarily descriptive but model-guided. The empirical work asks whether patterns in the data match the implications of endogenous product selection.

- **Firm-level comparisons.** Compare single- and multiple-product firms, and examine whether net adding/dropping is correlated with changes in firm outcomes.
- **Product-level analysis.** Examine whether products with high add rates also have high drop rates, and decompose product output by producer type.
- **Firm-product analysis.** Estimate whether a firm is more likely to drop a product when the product is smaller or newer within the firm.
- **Extensions.** Examine coproduction patterns and modes of product switching to identify facts not fully captured by the baseline model.

The product-drop regression is especially important:

$$Drop_{jpt} = \alpha_j + \alpha_p + \beta_1 \ln(Size_{jpt}) + \beta_2 \ln(Tenure_{jpt}) + \varepsilon_{jpt},$$

where j indexes firms and p indexes products. Firm fixed effects absorb all firm-level traits, and product fixed effects absorb product-level shocks. Identification comes from differences across products within the same firm and across firms within the same product.

7 Original figures and tables with reading notes

7.1 Table 1 – Prevalence of Firms Producing Multiple Products, Industries and Sectors in 1997; Table 2 – 1997 Multiple-Product versus Single-Product Firm Characteristics

TABLE 1—PREVALENCE OF FIRMS PRODUCING MULTIPLE PRODUCTS, INDUSTRIES AND SECTORS IN 1997

Type of firm	Percent of firms	Percent of output	Mean products, industries, or sectors per firm
Multiple product	39	87	3.5
Multiple-industry	28	81	2.8
Multiple-sector	10	66	2.3

Notes: The table categorizes firms according to whether they produce multiple products (five-digit SIC categories), industries (four-digit SIC categories), or sectors (two-digit SIC categories). Columns 1 and 2 summarize the distribution of firms and output, respectively. The final column reports the mean number of products, industries, and sectors across firms producing more than one of each.

Table 1 – Prevalence of Firms Producing Multiple Products, Industries and Sectors in 1997

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TABLE 2—1997 MULTIPLE-PRODUCT VERSUS SINGLE-PRODUCT FIRM CHARACTERISTICS

Firm characteristic	Multiple product	Multiple industry	Multiple sector
Output	0.66	0.67	0.92
Employment	0.58	0.61	0.86
Probability of export	0.12	0.12	0.16
Labor productivity	0.08	0.06	0.06
TFP	0.02	0.02	0.00

Notes: Results are from OLS regressions of log characteristics on a dummy variable indicating the firms' status as well as main industry fixed effects, i.e., the industry in which firms have the highest value of shipments. Regressions are restricted to the 110,414 observations for which all firm characteristics are available. All differences are statistically significant at the 1 percent level based on standard errors clustered by main industry except for multiple-sector firms' TFP.

Table 2 – 1997 Multiple-Product versus Single-Product Firm Characteristics

Read: Table 1 establishes the importance of multiple-product firms. Although they are a minority of firms, they account for most output. Multiple-sector firms are especially striking: only about 10% of firms operate across sectors, but they account for roughly two-thirds of manufacturing output.

Interpretation: Table 2 shows that multiple-product, multiple-industry, and multiple-sector firms are larger, more likely to export, and have higher revenue-based productivity than narrower firms. This matches the model's prediction that higher-productivity firms can cover more product fixed costs.

Takeaway: Product scope is not a side detail. It is closely linked to firm size, export status, and productivity.

7.2 Table 3 – Product Switching by U.S. Manufacturing Firms, 1987 to 1997; Table 4 – Product Switching and Changes in Firm Characteristics, 1987 to 1997

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TABLE 3—PRODUCT SWITCHING BY US MANUFACTURING FIRMS, 1987 TO 1997

Firm activity	All firms	Multi-product firms	Exporters	Large firms	Multi-plant firms
Panel A. Percent of firms					
None	46	20	38	39	25
Drop product(s) only	15	12	18	17	21
Add product(s) only	14	32	14	16	15
Both add and drop	25	36	31	28	38
Panel B. Output-weighted percent of firms					
None	11	6	6	10	5
Drop product(s) only	10	8	9	10	10
Add product(s) only	10	12	9	10	9
Both add and drop	68	75	76	70	77

Notes: Panel A displays average percent of surviving US manufacturing firms engaging in each type of product-chang-

Table 3 – Product Switching by U.S. Manufacturing Firms, 1987 to 1997

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TABLE 4—PRODUCT SWITCHING AND CHANGES IN FIRM CHARACTERISTICS, 1987 TO 1997

	Net drop	Net add	Observations	R^2
Log change in real output	−0.078***	0.096***	94,012	0.05
	(0.0093)	(0.0076)		
Log change in employment	−0.085***	0.078***	94,012	0.03
	(0.0100)	(0.0075)		
Log change in real output/worker	0.007**	0.018***	94,012	0.03
	(0.0038)	(0.0043)		
Change in TFP	−0.041***	0.031***	94,012	0.08
	(0.0070)	(0.0076)		

Notes: Table summarizes OLS regression results of log change in firm characteristics over five-year intervals according to whether firms net add or net drop products. Each row summarizes the regression for the noted dependent variable. Standard errors in parentheses are adjusted for clustering by product mix. Regressions include product mix by year fixed effects.
***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

Table 4 – Product Switching and Changes in Firm Characteristics, 1987 to 1997

Read: Table 3 is the paper's headline descriptive fact. About 54% of surviving manufacturing firms change their product mix over five years. Among multiple-product firms, product switching is even more common: only about 20% report no product change.

Interpretation: Output-weighted percentages make the pattern even stronger. Firms that change product mix account for nearly all manufacturing output, and firms that both add and drop products account for the largest output share. Table 4 then links net adding with growth in output, employment, labor productivity, and TFP; net dropping is associated with shrinking output, employment, and TFP.

Takeaway: Product switching is economically meaningful and systematically connected to firm growth and productivity dynamics.

7.3 Table 5 – Sector and Industry Switching by U.S. Manufacturing Firms, 1987 to 1997; Figure 1 – Average Product Add and Drop Rates, 1987 to 1997

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TABLE 5—SECTOR AND INDUSTRY SWITCHING BY US MANUFACTURING FIRMS, 1987 TO 1997

	Percent of firms		
Firm activity	Product activity	Industry activity	Sector activity
None	46	59	84
Drop only	15	14	6
Add only	14	13	6
Both add and drop	25	14	3

Notes: Table displays average share of surviving firms that engage in product, industry, and sector switching across five-year intervals from 1987 to 1997. Product, industry, and sector activity refers to the adding and/or dropping of five-digit, four-digit, and two-digit SIC cat-

Table 5 – Sector and Industry Switching by U.S. Manufacturing Firms, 1987 to 1997

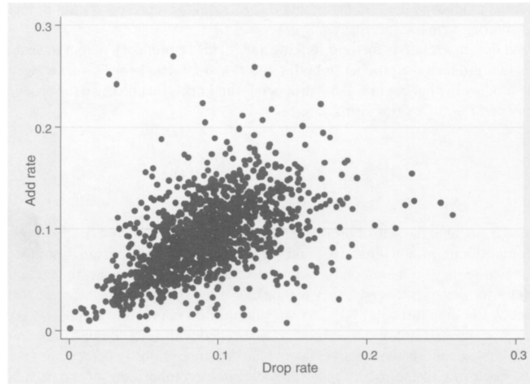

FIGURE 1. AVERAGE PRODUCT ADD AND DROP RATES, 1987 TO 1997

Note: Add (drop) rates are defined as the number of firms adding (dropping) the product between census years divided by the average number of firms producing the product in both years.

Figure 1 – Average Product Add and Drop Rates, 1987 to 1997

Read: Table 5 shows that product switching often crosses more aggregated boundaries. About 41% of firms change their four-digit industry set, and about 16% change their two-digit sector set.

Interpretation: Figure 1 shows a positive relationship between product add rates and product drop rates. Products that are frequently added are also frequently dropped. This pattern is difficult to explain by pure net reallocation from declining products to growing products; it is more consistent with turbulence and selection within product markets.

Takeaway: The product-switching margin is not just measurement noise. It changes firm scope at economically meaningful levels and reflects gross churn within products.

7.4 Table 6 – Decomposition of Product Output by Producer Type, 1987 to 1997; Table 7 – 1992 to 1997 Product Adding OLS Regressions

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TABLE 6—DECOMPOSITION OF PRODUCT OUTPUT BY PRODUCER TYPE, 1987 TO 1997

	Average Share (percent) of product output in year t produced by: Backward-looking			Forward-looking		
	Firms producing product in years $t-5$ and t (1)	Firms that add the product between years $t-5$ and t (2)	Firms born between years $t-5$ and t (3)	Firms producing product in years t and $t+5$ (4)	Firms that drop the product between years t and $t+5$ (5)	Firms that die between years t and $t+5$ (6)
Panel A						
1987	—	—	—	65	16	19
1992	67	14	19	67	15	18
1997	70	15	15	—	—	—
Panel B						
1987	—	—	—	44	27	29
1992	40	23	37	44	25	32
1997	45	26	29	—	—	—

Notes: The table reports average percentage decomposition of product output (panel A) and number of firms producing a product (panel B) according to firm activity. Columns 1–3 summarize backward-looking firm activities while columns 4–6 summarize forward-looking firm activities. Each row represents the average across all five-digit SIC products in the noted year. Decompositions cover the 1987 to 1997 censuses.

Table 6 – Decomposition of Product Output by Producer Type, 1987 to 1997

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TABLE 7—1992 TO 1997 PRODUCT ADDING OLS REGRESSIONS

	Single-product firms				Multiple-product firms			
	$Add_{t:t+5}$	$Add_{t:t+5}$	$Add_{t:t+5}$	$Add_{t:t+5}$	$Add_{t:t+5}$	$Add_{t:t+5}$	$Add_{t:t+5}$	$Add_{t:t+5}$
TFP_t	0.0140*** (0.0031)	0.0150*** (0.0031)			0.0020*** (0.0049)	0.0026*** (0.0049)		
$\ln(\text{output/worker})_t$			0.0099*** (0.0023)	0.0114*** (0.0023)			0.0178*** (0.0044)	0.0137*** (0.0044)
$\ln(\text{employment})_t$		0.0269*** (0.0012)		0.0271*** (0.0013)		0.0268*** (0.0022)		0.0265*** (0.0022)
$\ln(\text{age})_t$		0.0057*** (0.0021)		0.0053*** (0.0021)		0.0228*** (0.0036)		0.0224*** (0.0036)
R^2	0.07	0.07	0.07	0.07	0.51	0.52	0.51	0.52
Observations	105,035	105,035	105,035	105,035	74,976	74,976	74,976	74,976

Notes: Table summarizes OLS regression results of a dummy variable indicating product adding by single-product (left panel) and multiple-product (right panel) firms between years t and $t+5$ on year t covariates. Sample covers the five-year interval between 1992 and 1997. Regressions include product-mix and year fixed effects. Robust standard errors in parentheses are adjusted for clustering by product mix.

*** Significant at the 1 percent level.
** Significant at the 5 percent level.
* Significant at the 10 percent level.

Table 7 – 1992 to 1997 Product Adding OLS Regressions

Read: Table 6 decomposes product output into incumbents, adders, entrants, droppers, and exiters.

For the average product, newly adding firms and entering firms each account for sizable shares of output. Likewise, droppers and exiters both account for large forward-looking shares.

Interpretation: Table 7 shows that firms with higher initial revenue-based labor productivity or TFP are more likely to add products, for both single-product and multiple-product firms. This supports the model's selection mechanism: product adding is not random; more productive firms are better positioned to expand scope.

Takeaway: Product-level output dynamics combine entry/exit across firms with adding/dropping inside firms. Both margins matter.

7.5 Table 8 – 1992 to 1997 Firm-Product OLS Drop Regressions; Table 9 – Average Decomposition of Firm Output by Type of Product, 1987 to 1997

TABLE 8—1992 TO 1997 FIRM-PRODUCT OLS DROP REGRESSIONS

	$\text{Drop}_{t:t+5}$	$\text{Drop}_{t:t+5}$	$\text{Drop}_{t:t+5}$
ln(relative product size)$_t$	−0.059***	−0.086***	−0.077***
	(0.001)	(0.001)	(0.001)
ln(relative product tenure)$_t$	−0.189***	−0.219***	−0.223***
	(0.006)	(0.008)	(0.008)
Fixed effects	None	Firm	Firm, product
R^2	0.48	0.48	0.47
Observations	80,371	80,371	80,371

Notes: Table summarizes OLS regression results of a dummy variable indicating a firm-product drop between 1992 and 1997 on 1992 firm-product attributes and fixed effects. Firm-product size and tenure are relative to their average values across firms for the product in a given year. The regression sample is surviving multiple-product firms. Robust standard errors in parentheses are adjusted for clustering by product.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

Table 8 – 1992 to 1997 Firm-Product OLS Drop Regressions

TABLE 9—AVERAGE DECOMPOSITION OF FIRM OUTPUT BY TYPE OF PRODUCT, 1987 TO 1997

	Average share (percent) of firm output in year t accounted for by:			
	Backward-looking		Forward-looking	
	Products produced in years $t-5$ and t	Products added between years $t-5$ and t	Products produced in years t and $t+5$	Products dropped between years t and $t+5$
1987	—	—	71	29
1992	74	26	74	26
1997	69	31	—	—

Notes: Table reports the average percentage decomposition of firm output according to whether products were previously (left panel) or subsequently (right panel) produced. Each row represents the average across all surviving firms in the noted year.

Table 9 – Average Decomposition of Firm Output by Type of Product, 1987 to 1997

Read: Table 8 is central to the evidence for selection within firms. Firms are less likely to drop products with larger relative shipments and longer relative tenure. This result holds with firm fixed effects and product fixed effects.

Interpretation: Table 9 shows that recently added and soon-to-be-dropped products account for large shares of firm output. Product switching therefore reshapes the internal composition of surviving firms, not just a small fringe of their activity.

Takeaway: Firms reallocate resources away from smaller and newer product lines and toward products that survive and scale.

7.6 Table 10 – Mean Distribution of Within-Firm Output Shares, 1987 to 1997; Figure 2 – Distribution of Product Shipments within Firms, 1987 to 1997

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TABLE 10—MEAN DISTRIBUTION OF WITHIN-FIRM OUTPUT SHARES, 1987 TO 1997

Average percent of output	Number of products produced by the firm: 1	2	3	4	5	6	7	8	9	10
1	100	80	70	63	58	54	52	50	48	46
2		19	21	22	21	21	21	20	20	20
3			7	9	10	11	11	11	11	12
4				4	5	6	7	7	7	7
5					2	3	4	4	5	5
6						2	2	3	3	3
7							1	2	2	2
8								1	1	2
9									1	1
10										1

Notes: Columns indicate the number of products produced by the firm. Rows indicate the share of the products in firm output, in descending order of size. Each cell is the average across the relevant set of firm-products in the sample. Sample includes all firms producing at least ten products in the 1987 to 1997 censuses.

Table 10 – Mean Distribution of Within-Firm Output Shares, 1987 to 1997

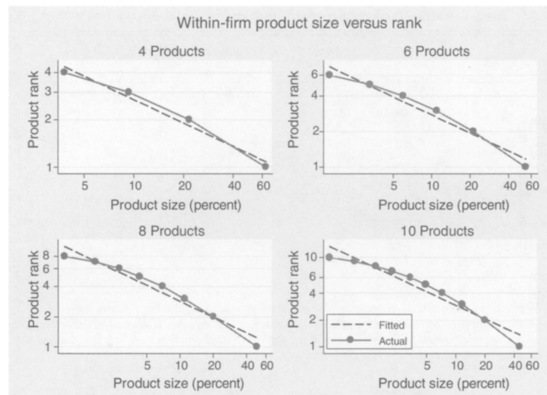

FIGURE 2. DISTRIBUTION OF PRODUCT SHIPMENTS WITHIN FIRMS, 1987 TO 1997

Notes: The solid line plots within-firm product rank against within-firm product size. Dashed lines are the result of an OLS regression of log product rank on log product size.

Figure 2 – Distribution of Product Shipments within Firms, 1987 to 1997

Read: Table 10 shows strong skewness within firms. Even among firms with many products, the largest product usually accounts for a large share of output. For example, among ten-product firms, the largest product accounts for nearly half of output on average.

Interpretation: Figure 2 compares the within-firm product-size distribution to a Pareto benchmark. Actual values deviate from the fitted line, with thinner tails than the Pareto distribution. The within-firm product distribution resembles the skewness studied in firm-size distributions, but it occurs inside firms.

Takeaway: Multiple-product firms are not balanced portfolios of equal product lines. They tend to have one or a few core products and a tail of smaller products.

7.7 Table 11 – Product Coproduction within Firms, 1987 to 1997; Table 12 – How Products are Added and Dropped, 1987 to 1997

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TABLE 11—PRODUCT COPRODUCTION WITHIN FIRMS, 1987 TO 1997

Sector	20	31	30	22	23	39	24	26	25	27	28	29	32	33	34	35	36	37	38
20 Food	158	2	9	5	16	5	6	9	6	11	42	3	4	6	15	16	8	5	4
31 Leather	2	1	1	1	3	1	1	1	1	1	1	0	0	1	2	1	1	0	0
30 Rubber and plastic	9	1	47	8	9	7	8	10	6	6	24	2	6	13	33	41	22	12	11
22 Textile	5	1	8	25	18	3	3	4	3	4	11	1	2	4	8	8	5	3	3
23 Apparel	16	3	9	18	64	6	6	6	6	9	12	1	3	7	20	21	9	5	5
39 Miscellaneous	5	1	7	3	6	10	3	3	2	9	6	0	1	3	10	9	6	3	3
24 Lumber	6	1	8	3	6	3	79	16	16	5	9	1	4	5	16	10	6	3	3
26 Paper	9	1	10	4	6	3	16	19	3	23	13	1	3	5	12	12	6	2	4
25 Furniture	6	1	6	3	6	2	16	3	23	3	4	0	1	4	12	10	7	3	3
27 Printing and pub.	11	1	6	4	9	9	5	23	3	401	10	1	2	5	13	15	10	4	5
28 Chemicals	42	1	24	11	12	6	9	13	4	10	74	13	10	16	25	29	20	10	18
29 Petroleum	3	0	2	1	1	0	1	1	0	1	13	4	2	3	4	3	2	1	1
32 Stone and concrete	4	0	6	2	3	1	4	3	1	2	10	2	10	6	11	11	7	3	3
33 Primary metal	6	1	13	4	7	3	5	5	4	5	16	3	6	27	48	47	29	15	10
34 Fabricated metal	15	2	33	8	20	10	16	12	12	13	25	4	11	48	124	117	53	30	22
35 Industrial mach	16	1	41	8	21	9	10	12	10	15	29	3	11	47	117	119	72	38	31
36 Electronic	8	1	22	5	9	6	6	6	7	10	20	2	7	29	53	72	70	25	37
37 Transportation	5	0	12	3	5	3	3	2	3	4	10	1	3	15	30	38	25	18	10
38 Instruments	4	0	11	3	5	3	3	4	3	5	18	1	3	10	22	31	37	10	14

Table 11 – Product Coproduction within Firms, 1987 to 1997

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TABLE 12—HOW PRODUCTS ARE ADDED AND DROPPED, 1987 TO 1997

	Share of products		Share of firms	
	Unweighted	Value-weighted	Unweighted	Value-weighted
Panel A. Method of product adding				
Existing plant(s) only	0.862	0.412	0.899	0.259
Acquired plant(s) only	0.055	0.259	0.013	0.048
New plant(s) only	0.060	0.120	0.030	0.044
Combination with M&A	0.013	0.147	0.031	0.488
Combination without M&A	0.011	0.061	0.027	0.161
Panel B. Method of product dropping				
Existing plant(s) only	0.835	0.320	0.898	0.213
Divested plant(s) only	0.052	0.282	0.007	0.018
Closed plants only	0.084	0.123	0.037	0.042
Combination with M&A	0.015	0.231	0.026	0.538
Combination without M&A	0.013	0.045	0.033	0.189

Notes: Table reports the manner in which firms add (panel A) and drop (panel B) five-digit SIC products. The first two columns report the distribution with respect to products; the second two columns report the distribution with respect to firms. Figures shown are averages across the pooled 1987 to 1997 sample.

Table 12 – How Products are Added and Dropped, 1987 to 1997

Read: Table 11 shows that some sector pairs are coproduced much more often than random assignment would imply. Coproduction is especially common within related sectors and along the diagonal.

Interpretation: Table 12 shows how firms add and drop products. Most product adds and drops happen within existing plants. M&A accounts for a small share of the number of product switches but a much larger share of the value of products added and dropped.

Takeaway: The baseline model captures selection well, but Table 11 and Table 12 point to richer forces: shared capabilities, product complementarities, plant-level reorganization, and acquisitions/divestitures.

8 How the evidence maps into the model

- **Firm productivity and scope.** Tables 1 and 2 support the model's implication that more productive firms have wider product scope.
- **Firm-level shocks.** Table 4 supports the idea that net product adding accompanies firm expansion and productivity growth, while net dropping accompanies contraction.
- **Product-market turbulence.** Figure 1 supports the model's product-level implication that products with high add rates also have high drop rates.
- **Firm-product selection.** Table 8 supports the model's strongest firm-product prediction: small and young firm-product pairs are more likely to be dropped.

- **Intensive versus extensive margin.** Table 10 and Figure 2 show that product scope alone is not enough to understand firm size. Intensive output per product is very important.
- **Beyond the baseline model.** Tables 11 and 12 show that product relatedness and ownership changes matter. The baseline model abstracts from these forces but the paper uses them to motivate extensions.

9 Economic interpretation

The paper changes the empirical picture of industry dynamics. Standard reallocation studies ask whether productive firms replace unproductive firms. Bernard, Redding, and Schott add a new question: do firms themselves reallocate resources across products as they learn which product lines are most profitable?

The answer is yes. Product switching is frequent, pervasive, and large. The core mechanism is selection. A firm can be productive overall but still have weak products; it can drop these products while adding others. This means productivity growth can come not only from firm turnover or growth of productive firms, but also from internal reallocation across product lines.

10 Limitations and caveats

- **Five-year frequency.** The quinquennial census interval misses product additions and drops that occur and reverse within five years.
- **Product aggregation.** Five-digit SIC codes are broad compared with true products or SKUs. Product switching at finer levels is likely even more frequent.
- **Revenue productivity.** The paper generally observes revenue-based productivity, not physical productivity. Price and demand differences can affect measured productivity.
- **No product-level input data.** The census does not provide inputs by firm-product, making product-level productivity measurement impossible.
- **Model abstraction.** The baseline model abstracts from product complementarities, shared production capabilities, and M&A, even though the data show these are relevant.

11 Bottom line

- Product switching is a central dimension of firm dynamics.
- Multiple-product firms dominate U.S. manufacturing output.
- More productive firms are more likely to add products and produce wider product ranges.
- Product dropping is systematic: smaller and newer products are more likely to be dropped.
- Gross product adding and dropping are large relative to net changes.
- The paper's key contribution is to show that resource reallocation occurs within firms, not only across firms.